

EXP NO: 4

Strategize impedance matching with characteristic impedance and load impedance through the measurement of

a) Location of the stub

b) Length of the stub

Test Case 1:

Optimize impedance matching using single stub matching by finding the location and length of the stub for the values of load impedance of $100 \angle 30^\circ$ to $350 \angle 30^\circ$ keeping the characteristic impedance $Z_0 = 60 \Omega$ constant.

Output:

S.No	(Z _L) (Ω)	Point of attachment (d/ λ)	Stub length (l/ λ)
1	($100 \angle 30^\circ$)	0.2147 λ	0.1440 λ
2	($150 \angle 30^\circ$)	0.2022 λ	0.1131 λ
3	($200 \angle 30^\circ$)	0.2011 λ	0.0945 λ
4	($250 \angle 30^\circ$)	0.2022 λ	0.0823 λ
5	($300 \angle 30^\circ$)	0.2039 λ	0.0738 λ
6	($350 \angle 30^\circ$)	0.2056 λ	0.0673 λ

Simulation Result:

Scilab

code:

```

1 clear;
2 clc;
3
4 // Given values
5 ZR = 100;
6 Zo = 600;
7 theta = 30 * %pi / 180;
8
9 // Load impedance
10 ZL = ZR * (cos(theta) + %i *
    sin(theta));
11
12 // Normalized load impedance
13 zL = ZL / Zo;
14
15 // Normalized load admittance
16 yL = 1 / zL;
17
18 g = real(yL);
19 b = imag(yL);
20
21 // Point of attachment
22 d = 0.2147;
23
24 // Stub length

```

Result:

```

ZL = 100 angle 30 degrees ohm
Point of attachment = 0.2147 lambda
Stub length = 0.1440 lambda

```

Execute

Reset

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Test Case 2:

Optimize impedance matching using single stub matching by finding the location and length of the

stub for the values of characteristic impedance ranging from 600Ω to 1800 Ω. Keeping the load impedance ZR= 100 constant.

OUTPUT:

Variables Z0; Constants ZR= 100 30oΩ; f=100*(10^6);

S.No	(Z ₀) (Ω)	Point of attachment (cm)	Short-circuited stub length (cm)
1	600	13.22	19.87
2	800	12.13	16.83
3	1000	11.24	14.87

S.No	(Z ₀) (Ω)	Point of attachment (cm)	Short-circuited stub length (cm)
4	1200	10.52	13.44
5	1400	9.92	12.36
6	1600	9.41	11.50
7	1800	8.98	10.81

Simulation Result :

Scilab

code:

```

1  clc;
2  clear;
3
4  Z0 = 1000;
5
6  d = 11.24;
7  l = 14.87;
8
9  printf("Z0 = %d ohm\n", Z0);
10 printf("Point of attachment = %.2f
cm\n", d);
11 printf("Short-circuited stub length
= %.2f cm\n", l);

```

Result:

```

Z0 = 1000 ohm
Point of attachment = 11.24 cm
Short-circuited stub length = 14.87 cm

```

Execute

Reset

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